

Makefile basics

Make reads a Makefile and runs named targets

Targets, deps, and .PHONY

- A rule is a target, optional dependencies, then a tab-indented recipe
- Make rebuilds when the target is missing or older than a dependency
- Names like clean are usually marked .PHONY so Make always runs them even if a file with
- Default make often means all or the first target, here, build

Browser lab

Digital Design and Verification Platform

HomeToolsMakefile

Learn **targets**, **deps**, **variables**, and **.PHONY** on a tiny chip Makefile — run Make and watch what rebuilds.

Starter example: Chip Makefile with `sim`, `build/$(TB)`, `vp`, `deps`, and `.PHONY: sim clean`. Run `make sim`, touch `RTL_`, run again — see the rebuild.

Load starter example

Challenges 0 / 22 cleared

Quiz: target: Left of the colon is the? Answer: target

Answer

Show hintCheckNextSkip

Quiz: target

Quiz: deps

Quiz: phony

Quiz: expand

First sim

Up to date compile

Touch rebuilds

Clean

TB override

Quiz: recipe

Quiz: ?*

Quiz: why phony

Rebuild after clean

Quiz: order

Both vgs

Quiz: build dir

Touch TB

Starter missing out

Quiz: SIM

Not up-to-date after clean

Quiz: default TB

Pull flow

Core ideas

target: deps

Recipe runs if target is missing or older than any dependency.

.PHONY

Marks names that are not real files — always run if requested.

Makefile

```
$(RM)iverilog -TbTb_alu -clock13

# this project Makefile (lab model)
RM ?= iverilog
TB ?= Tb_alu
RTL = rtl_alu.v

.PHONY: sim clean

build:
    mkdir -p build

build/$(TB).vvp: $(RTL) tb/$(TB).v | build
    $(RM) -rf build
    $(RM) -rf $(TB)
    $(RM) -rf $(TB)_top.v

sim: build/$(TB).vvp
    vvp build/$(TB).vvp

clean:
    rm -rf build
```

Filesystem & Make

default goal (aka: `sim` - current TB product missing)

```
rtl_alu.v | tb/Tb_alu.v | tb/Tb_top.v
tb/Tb_alu.v | tb/Tb_top.v | tb/Tb_top.v
tb/Tb_top.v | tb/Tb_top.v | tb/Tb_top.v
tb/Tb_top.v | tb/Tb_top.v | tb/Tb_top.v
build/rtl_alu.vvp | tb/Tb_top.v | tb/Tb_top.v
build/rtl_alu.vvp | tb/Tb_top.v | tb/Tb_top.v
```

Touch (bump md5)

rtl_alu.v | tb/Tb_alu.v | tb/Tb_top.v

make sim

make sim TbTb_top

make clean

make sim (again / up-to-date?)

Make output

```
# starter tree: sources present, build/ empty
```

Cheat sheet

Place	Meaning
target: dep1 dep2	Build target from deps
Tab-indented lines	Recipe shell commands
VARIABLE ?= value	Default assign (override from CLI)
\$(VARIABLE)	Expand variable
.PHONY: sim clean	Not real files — always eligible

- if Make prints nothing useful and says up to date, nothing was older.
- Order only deps (after `||`) must exist but don't force rebuild on timestamp — lab uses `build/` as a dir gate.
- Always declare phony targets like `sim/clean`, so a file named `clean` cannot block them.

Makefile basics lab

learn_unix — make basics

Real shell practice

```
wsl - module24-make-basics/examples/makefile_basics/sample_project

# Track A - real Unix
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module24-make-basics/examples/makefile_basics/sample_project

$ ls -la
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu 592 Jul 18 16:34 Makefile
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 build
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 logs
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 src
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 tb

$ make help
Targets: all (default), build, run, clean, help

$ make
[Make] Creating build/ and logs/...
[Make] Done.

$ make run
[Make] Creating build/ and logs/...
[Make] Done.
[Make] Running demo ...
[INFO] Project: /mnt/d/proj/designs/learning/courses/learn_unix/module24-make-
basics/examples/makefile_basics/sample_project
[Make] Done. See logs/run.log

$ ls -la build logs
build:
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu  0 Jul 20  2026 .gitkeep

logs:
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20  2026 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu  0 Jul 18 16:34 .gitkeep
-rwxrwxrwx 1 yongfu yongfu 125 Jul 20  2026 run.log

$ make clean
[Make] Cleaning build/ and logs/...
[Make] Done.

$ ls build logs
build:

logs:
```

Real shell — make targets

Real shell practice — try these

```
# pwd — print working directory (where am I?)  
pwd
```

```
# ls -la — list all entries, long format (what is here?)  
ls -la
```

```
# make help — list available targets  
make help
```

```
# make — default target (all → build); creates build/ and logs/  
make
```

```
# make run — depends on build; appends a line to logs/run.log  
make run
```

```
# ls -la build logs — confirm generated dirs and the run log
```

Pitfalls to watch

- Run make from the project root so paths in the Makefile resolve
- Recipes must use a real tab, not spaces
- Do not create a file named clean if clean is a phony target, or mark it .PHONY
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, clear a few challenges after the starter
- On the real shell, run `help`, `build`, `run`, and `clean` in the sample project
- When you are ready, take the short quiz, then continue to the dry-run mindset module